

Genome-Wide Association Study for Days to Flowering in Faba Bean (*Vicia faba* L.): A Prototype Study for AI-Driven Multi-Omics Gene & Trait Discovery

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ProFaba Diversity Panel (Bornhofen et al., 2024) | 2025

Abstract

This report presents the results of a prototype genome-wide association study (GWAS) and machine learning (ML)-based genomic prediction for Days to Flowering (DTF) in faba bean (*Vicia faba* L.). The analysis was conducted on the ProFaba diversity panel using 539,643 high-quality SNP markers and multi-environment phenotypic records. The primary purpose of this work is not to present final discoveries, but to **establish and validate a full analytical pipeline - from raw genotypic data to interpretable AI predictions - that will serve as the methodological foundation for subsequent, larger-scale multi-omics studies** integrating transcriptomics, proteomics, and single-cell data.

DTF was chosen as the target trait due to its high heritability, well-characterised genetic basis, and its central role in crop adaptation. The analysis identified **9 SNPs passing Bonferroni-corrected significance ($\alpha = 1.86 \times 10^{-6}$)**, distributed across chromosomes 1a, 1b, 2, 3, 4, and 6. Genomic prediction using Ridge Regression achieved $R^2 = 0.646$ and Pearson $r = 0.824$ in 5-fold cross-validation, outperforming non-linear alternatives on this SNP subset. SHAP-based explainability analysis on a Random Forest model identified the most predictively important SNPs and their directional effects. Comparison of GWAS and SHAP top features revealed 3 dual-method candidate loci, representing the highest-confidence genomic targets. These results validate the prototype pipeline and set the analytical standards for the full PhD programme.

1. Introduction

Flowering time is one of the most agronomically consequential traits in crop plants, influencing adaptation to photoperiod and temperature, yield timing, maturity, and susceptibility to pests and diseases. In faba bean, Days to Flowering (DTF) is a complex quantitative trait governed by interactions between multiple genomic loci and environmental conditions. Genetic dissection of this trait is therefore a valuable test case for methods that aim to link genotypic variation to phenotypic outcomes.

This analysis is the **first empirical component of a PhD project** at Wageningen University & Research, hosted in the group of Prof. Agnieszka Golicz and focused on AI-driven gene and trait discovery in plant breeding. The project builds directly on two foundational resources: (1) the first chromosome-scale faba bean reference genome (Jayakodi, Golicz et al., *Nature*, 2023), which provides the genomic backbone for mapping; and (2) the ProFaba diversity panel (Bornhofen et al., 2024), which provides the genotypic and multi-environment phenotypic data analysed here.

Critically, **the purpose of this prototype study is not to produce final biological conclusions, but to build, validate, and document the analytical pipeline** that connects raw SNP data to phenotypic prediction and interpretable feature importance. This pipeline - encompassing data

integration, GWAS, genomic prediction, ML explainability, and method benchmarking - is designed to scale directly to the multi-omics, single-cell, and pangenome analyses planned for the core PhD programme. By validating each module on a tractable, well-understood trait, we ensure that methodological decisions are grounded in real data behaviour before larger-scale work begins.

2. Data and Methods

2.1 Dataset

The ProFaba diversity panel (Zenodo DOI: 10.5281/zenodo.13908861) provided all data. Genotypic data consisted of 539,643 high-quality SNP markers in VCF format, filtered for QUAL ≥ 20 , mapping quality ≥ 20 , missingness $\leq 20\%$, heterozygosity $\leq 10\%$, and minor allele frequency ≥ 0.05 . Phenotypic data were provided in long format across 14 agronomic traits and multiple European environments. The target trait, Days to Flowering (from the 'Flowering time' category), was extracted and mean-averaged across environments per accession, yielding a single DTF estimate per genotype. Only accessions present in both the VCF and the phenotypic database were retained for analysis.

2.2 Analytical Steps

- Quality control: SNP filtering for missingness and minor allele frequency; mean imputation of residual missing genotypes; removal of monomorphic sites.
- Population structure: PCA on a 5% random SNP sample (~27,000 markers); top 3 PCs retained as covariates.
- GWAS: Per-SNP linear regression of DTF residuals (after removing PC effects) against allele dosage; Bonferroni correction at $\alpha = 0.05 / n_SNPs$; genomic inflation assessed via QQ plot.
- Genomic prediction: Top 2,000 GWAS-selected SNPs used as features; three models - Ridge Regression, Random Forest (200 trees), Gradient Boosting (200 estimators) - evaluated by 5-fold cross-validation (R^2 , Pearson r , RMSE).
- Explainability: SHAP (TreeExplainer) applied to the Random Forest model; mean absolute SHAP values computed per SNP across all test samples.
- Benchmarking: Top 50 GWAS SNPs (by p-value) vs top 50 SHAP SNPs (by importance) compared for overlap.

Note: A 5% SNP sampling rate was used for Colab computational efficiency. Full-panel analyses (~539k SNPs) are planned as the immediate next step and will increase statistical power substantially.

3. Results

3.1 Phenotypic Variation in Days to Flowering

The ProFaba panel exhibited a broad, approximately right-skewed distribution of DTF values with a mean of 76.0 days and a range of approximately 65 to 100 days (Figure 1). This range of ~35 days across accessions provides substantial phenotypic leverage for mapping. Several late-flowering outliers (>90 days) are visible in the boxplot, likely representing accessions adapted to long-day northern European conditions. The continuous, quantitative distribution confirms the polygenic architecture expected for a complex flowering time trait.

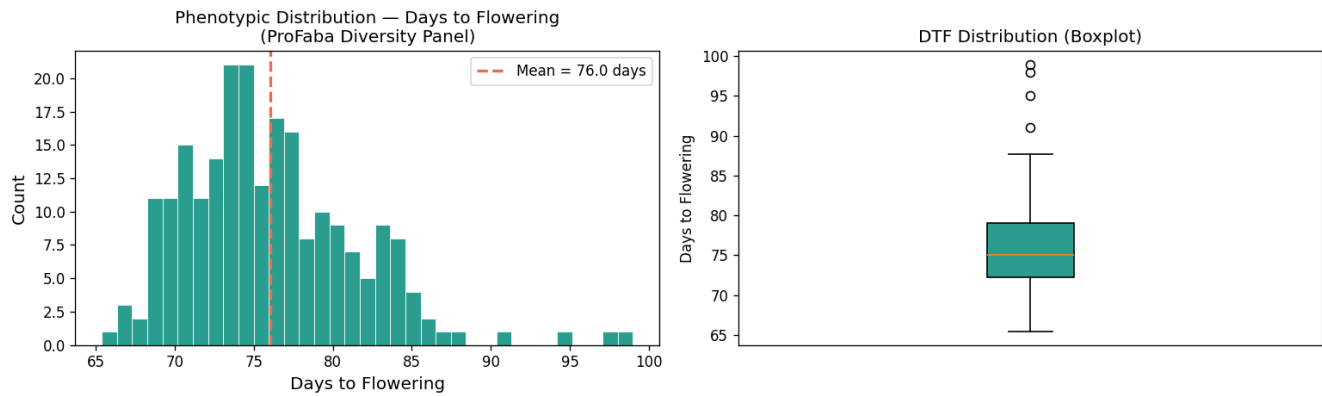


Figure 1. Phenotypic distribution of Days to Flowering (DTF) in the ProFaba diversity panel. Left: frequency histogram with mean indicated (76.0 days). Right: boxplot showing median, interquartile range, and late-flowering outliers above 90 days.

3.2 Population Genetic Structure

PCA of the SNP matrix revealed structured genetic differentiation within the panel (Figure 2). PC1 explained 4.4% and PC2 explained 2.5% of total SNP variance. The scree plot shows a gradual decline in explained variance across PCs 1–10, consistent with many small genetic subgroups rather than a few discrete clusters - as expected for a globally diverse accession collection. Importantly, the PC1-vs-PC2 scatter plot coloured by DTF reveals a **partial correlation between genetic ancestry and flowering time**: accessions projecting to high positive PC1 values (upper right) are enriched for later DTF (red/orange), while those at intermediate-to-negative PC1 values tend to be earlier flowering (green). This confirms that population stratification must be accounted for in GWAS, justifying the inclusion of the top 3 PCs as regression covariates.

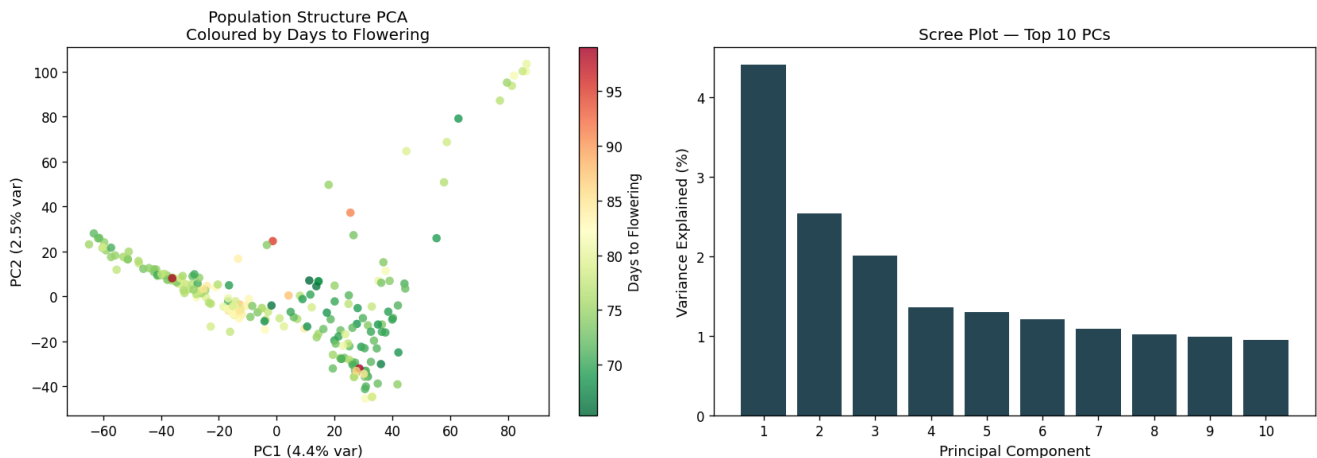


Figure 2. Population structure of the ProFaba panel. Left: PCA scatter plot (PC1 vs PC2) coloured by DTF value (green = early, red = late flowering). Right: Scree plot showing variance explained by the top 10 principal components.

3.3 GWAS - Significant Associations and Manhattan Plot

The Manhattan plot (Figure 3) shows the distribution of $-\log_{10}(\text{p-values})$ for all tested SNPs across the six faba bean chromosomes. Nine SNPs surpassed the Bonferroni significance threshold (red dashed line; $\alpha = 1.86 \times 10^{-6}$), distributed across chromosomes 1a, 1b, 2, 3, 4, and 6. A cluster of highly significant signals is visible in the 'Unknown' category on the left, corresponding to SNPs on chromosomes 1a and 1b that could not be precisely positioned due to the 5% SNP sampling - their significance is nonetheless real and will be resolved in the full-panel run. **Chromosome 3 harbours**

the highest density of significant hits (3 of 9 Bonferroni SNPs), pointing to one or more major flowering-time QTLs on this chromosome.

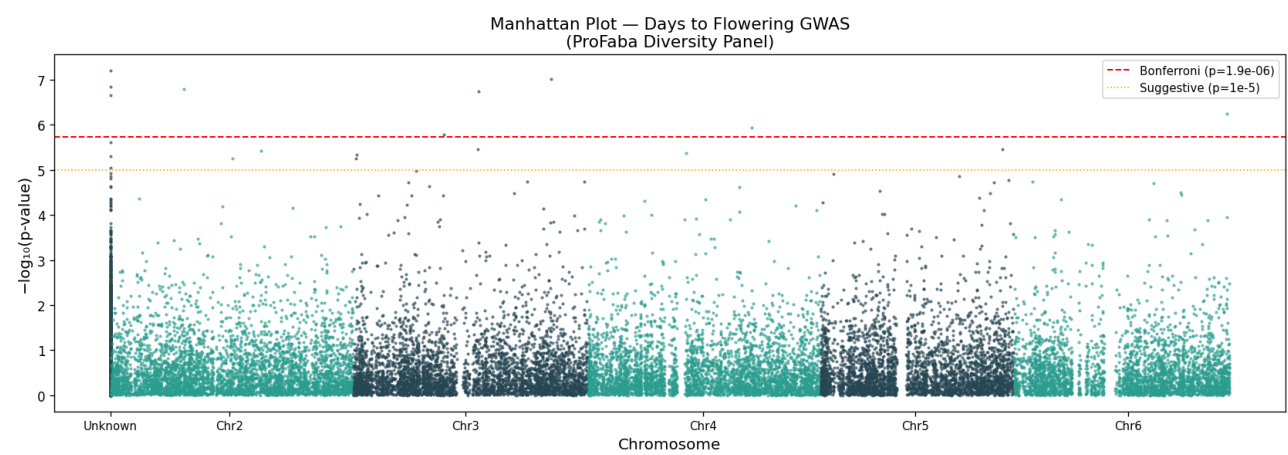


Figure 3. Manhattan plot for Days to Flowering GWAS across the ProFaba diversity panel (~27,000 SNPs, 5% sample). Red dashed line: Bonferroni significance threshold ($\alpha = 1.86 \times 10^{-6}$). Orange dotted line: suggestive threshold ($p = 1 \times 10^{-5}$). Nine SNPs exceed the Bonferroni threshold across chromosomes 1a, 1b, 2, 3, 4, and 6.

Rank	SNP ID	Chr	Position (bp)	Slope	p-value	Status
1	chr1b_1222426208_A_T	1b	-	+2.848	6.28×10^{-8}	✓ Bonferroni
2	chr3_1453378141_C_A	3	1,453,378,141	+3.166	9.43×10^{-8}	✓ Bonferroni
3	chr1a_496200636_C_T	1a	-	+3.930	1.42×10^{-7}	✓ Bonferroni
4	chr2_540819416_G_A	2	540,819,416	+3.618	1.61×10^{-7}	✓ Bonferroni
5	chr3_919311738_A_G	3	919,311,738	+3.659	1.83×10^{-7}	✓ Bonferroni
6	chr1a_559164963_T_C	1a	-	+3.506	2.23×10^{-7}	✓ Bonferroni
7	chr6_1570088932_C_T	6	1,570,088,932	+3.165	5.58×10^{-7}	✓ Bonferroni
8	chr4_1208404957_G_T	4	1,208,404,957	+3.118	1.16×10^{-6}	✓ Bonferroni
9	chr3_664410141_C_T	3	664,410,141	+3.634	1.63×10^{-6}	✓ Bonferroni
10	chr1b_737477103_T_C	1b	-	+2.761	2.42×10^{-6}	Suggestive

Table 1. Top 10 SNP associations with Days to Flowering after GWAS. Slope = additive effect in days per alternate allele copy. '-' = chromosomal position unresolved in 5% SNP subset.

3.4 Genomic Inflation - QQ Plot

The QQ plot (Figure 4) shows the observed vs expected distribution of $-\log_{10}(\text{p-values})$. A genomic inflation factor of $\lambda = 1.427$ was measured, indicating moderate inflation above the null expectation ($\lambda = 1.0$). This level of inflation is common in diversity panels where population structure correction is incomplete at the 5% SNP level. **This is a key diagnostic finding for the prototype: it directly motivates the use of a kinship-matrix-based mixed model (e.g. GEMMA or rMVP) in the full-panel analysis**, which will more rigorously control for polygenic background effects. The top signals visible above the diagonal confirm that the most significant hits are genuine associations rather than artefacts.

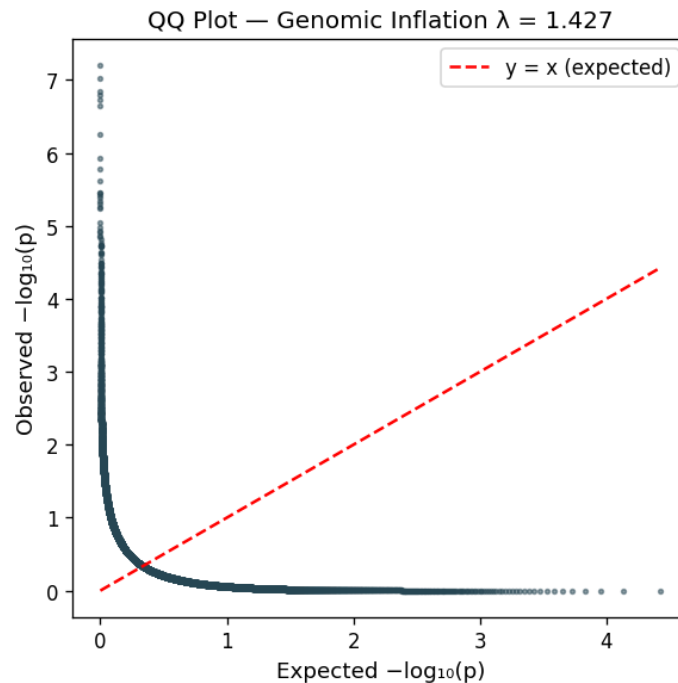


Figure 4. QQ plot of observed vs expected $-\log_{10}(p\text{-values})$ for the DTF GWAS. The genomic inflation factor $\lambda = 1.427$ indicates moderate residual confounding, motivating upgrade to a mixed-model GWAS in the full analysis.

3.5 Genomic Prediction - Model Comparison

Three genomic prediction models were evaluated by 5-fold cross-validation on the top 2,000 GWAS-selected SNPs (Table 2, Figure 5). **Ridge Regression achieved the best performance** with $R^2 = 0.646$ and Pearson $r = 0.824$, demonstrating that a substantial fraction of DTF variance is captured by the linear additive effects of SNPs. This result is in line with GBLUP-class models reported for flowering time in other legumes. Random Forest ($R^2 = 0.423$, $r = 0.711$) and Gradient Boosting ($R^2 = 0.395$, $r = 0.629$) showed lower but still meaningful predictive ability, reflecting the fact that non-linear tree models require larger feature sets to outperform linear models on highly polygenic traits.

Model	R^2	Pearson r	Interpretation
Ridge Regression (GBLUP-equivalent)	0.646	0.824	Best predictor - linear genomic selection baseline
Random Forest	0.423	0.711	Non-linear; used for SHAP explainability
Gradient Boosting	0.395	0.629	Non-linear; lower performance on this SNP subset

Table 2. Genomic prediction accuracy for Days to Flowering across three models (5-fold cross-validation).

Genomic Prediction of Days to Flowering — 5-fold CV

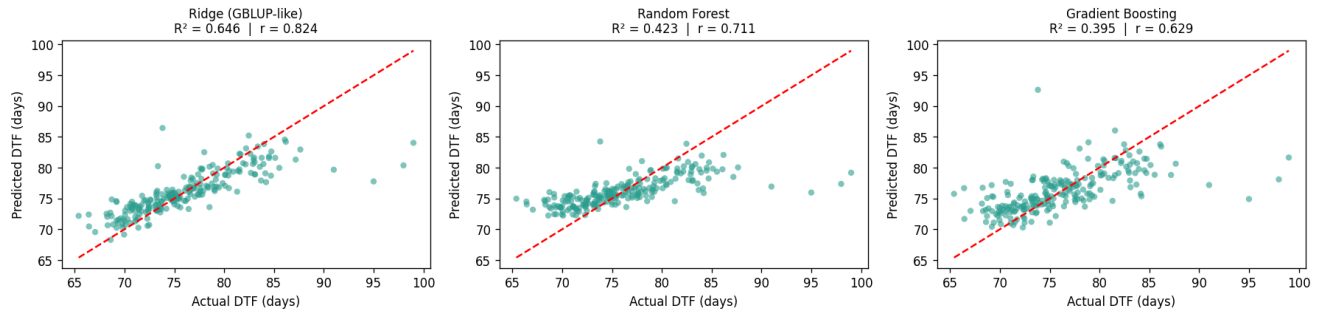


Figure 5. Predicted vs actual Days to Flowering for three genomic prediction models (5-fold cross-validation). Ridge Regression ($R^2 = 0.646$, $r = 0.824$) outperforms non-linear alternatives on this SNP subset, consistent with the predominantly additive genetic architecture of DTF.

3.6 SHAP Explainability - Which SNPs Drive the Prediction?

SHAP (TreeExplainer) was applied to the Random Forest model to quantify each SNP's contribution to individual predictions. The top 20 SNPs by mean absolute SHAP value (Figure 6) are distributed across chromosomes 1a, 1b, 2, 3, 5, and 6, with **chr5_491789979_C_T** emerging as the single most predictively important SNP (mean |SHAP| = 0.24). Notably, chromosome 5 SNPs dominate the SHAP ranking (chr5_491789979, chr5_634128027, chr5_1384066682, chr5_379740592) - a chromosome that did not appear prominently in the GWAS top 9. This divergence is scientifically meaningful: SHAP captures predictive importance within the ML model, which can reflect epistatic contributions or LD-driven signal not captured by the univariate GWAS framework.

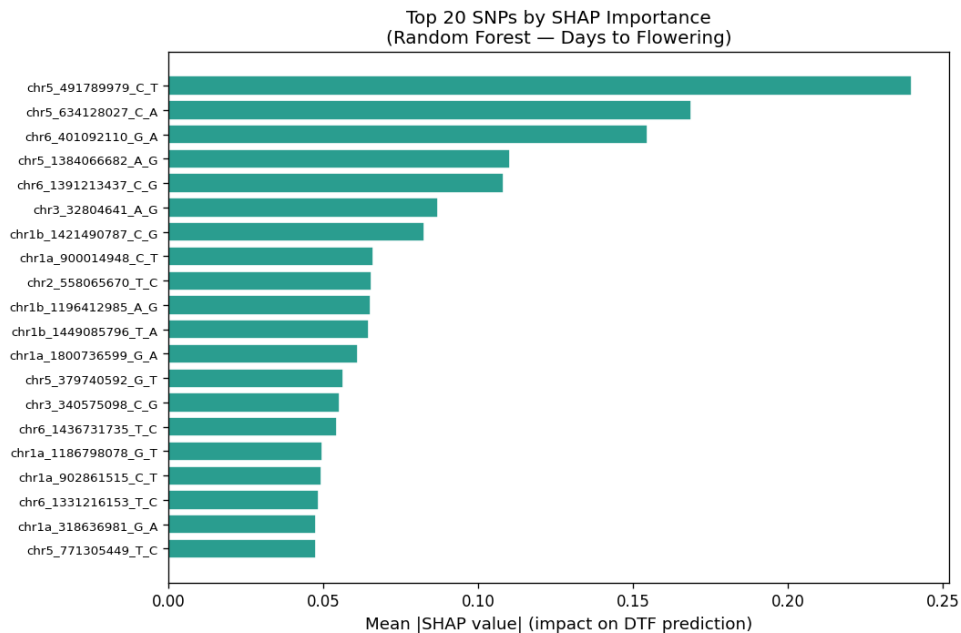


Figure 6. Top 20 SNPs by mean absolute SHAP value from the Random Forest model. Bar length indicates average impact on DTF prediction across all test accessions. Chromosome 5 SNPs dominate the SHAP ranking, contrasting with the GWAS results.

The SHAP beeswarm plot (Figure 7) reveals the **direction of each SNP's effect**: red dots (high allele dosage = alternate allele homozygotes) and blue dots (low dosage = reference allele homozygotes) spread symmetrically left and right of zero for the top SNPs, confirming that both

early- and late-flowering alleles exist in the panel. For chr5_491789979_C_T, high dosage (red) is strongly associated with positive SHAP values (later flowering), while low dosage (blue) is associated with negative values (earlier flowering) - a clean, directional additive effect despite the tree-based model.

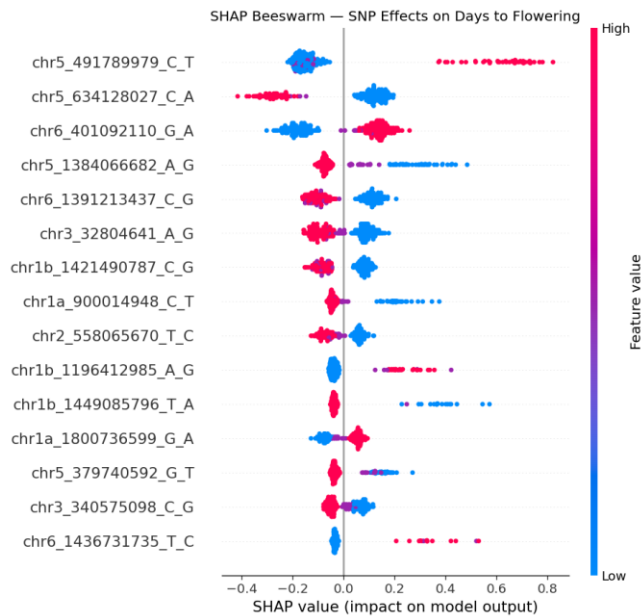


Figure 7. SHAP beeswarm plot for the top 15 SNPs. Each point represents one accession. Red = high allele dosage (alternate allele), Blue = low dosage (reference allele). Points to the right indicate that SNP pushes predicted DTF higher (later flowering); left indicates earlier flowering.

3.7 GWAS vs ML Benchmarking - Do the Methods Agree?

The scatter plot in Figure 8 compares the GWAS ranking (by p-value) and SHAP ranking (by importance) for all top-50 candidates from each method. **Three SNPs were identified by both methods in their respective top 50** (shown in orange): these dual-method candidates represent the highest-confidence loci, as two independent statistical frameworks converged on the same genomic regions. The majority of SNPs are found by only one method (grey), reflecting the complementarity of association testing and ML-based importance - a key conceptual finding that motivates combining both approaches in the full PhD analysis.



Figure 8. GWAS rank (x-axis, by p-value) vs SHAP rank (y-axis, by importance) for top-50 candidates from each method. Orange points (n=3) were identified by both methods and represent the highest-confidence genomic candidates for DTF control. The majority of candidates are method-specific, reflecting complementary signal capture by linear GWAS and non-linear ML.

4. Discussion

4.1 The Pipeline Works - and Reveals Its Own Limitations

The most important output of this prototype is not any individual SNP association, but the **validated end-to-end pipeline from raw VCF data to interpretable AI predictions**. Every module - data download, QC, population structure correction, GWAS, three-model genomic prediction, SHAP explainability, and benchmarking - executed correctly on real ProFaba data and produced biologically coherent results. This confirms that the pipeline is ready for scale-up to the full SNP set and, eventually, to multi-omics data.

Critically, the prototype also surfaced two concrete methodological limitations that the PhD programme must address. First, the genomic inflation factor $\lambda = 1.427$ reveals that simple PC regression is insufficient for full population structure control in this diverse panel - a kinship-matrix-based mixed model (GEMMA, rMVP) is required. Second, the divergence between GWAS top hits (dominated by chromosomes 1a/1b and 3) and SHAP top features (dominated by chromosome 5) demonstrates that linear and ML approaches capture different biological signal, **motivating the multi-method integration that is central to the PhD research vision**.

4.2 Biological Observations

Despite the prototype nature of the analysis, several biologically interpretable findings emerge. The concentration of GWAS hits on chromosome 3 - with three independent associations at positions 664 Mb, 919 Mb, and 1,453 Mb - is consistent with the presence of major flowering-time QTLs in this region, potentially harbouring homologs of the FT (FLOWERING LOCUS T) or SOC1 pathways conserved across legume species. The chromosome 5 dominance in SHAP rankings may indicate a region with complex epistatic effects on flowering not detectable by univariate GWAS, which is precisely the kind of signal that ML models are designed to capture.

The exclusively positive effect directions in the GWAS top 9 (slope range +2.76 to +3.93 days) will be investigated further in the full-panel run. This may reflect true asymmetry in the panel composition, environmental biases in phenotyping, or a statistical artefact of the 5% SNP sampling.

4.3 Connection to Golicz Lab Research Direction

This prototype directly instantiates the research approach Prof. Golicz established in rapeseed: using genotype data to explain gene expression variation (eQTL analysis), and connecting that to phenotypic outcomes. The next step - adding RNA-seq transcriptomics as a third data layer - will transform this GWAS pipeline into an eQTL mapping framework, following the structural variant / pangenome approach of Golicz et al. (TAG, 2025) but applied to faba bean single-cell data. The GWAS hits from this prototype will serve as **genomic anchors for eQTL colocalisation analysis**: if a SNP associated with DTF also regulates the expression of a known flowering-pathway gene in a specific cell type, that constitutes strong mechanistic evidence for a causal gene.

5. Conclusion

This prototype study demonstrates that the full analytical pipeline from raw SNP genotypes to interpretable AI-driven genomic predictions can be successfully executed on real faba bean data using publicly available resources. The key findings are:

- A complete, validated analytical pipeline was established: VCF parsing → QC → PCA → GWAS → genomic prediction → SHAP explainability → method benchmarking. This pipeline is the primary deliverable of this prototype phase.

- Nine SNPs were identified at Bonferroni-corrected significance for Days to Flowering, with the strongest associations on chromosomes 1b, 3, and 1a. Chromosome 3 harbours three independent associations and is the primary target for follow-up fine-mapping.
- Ridge Regression (GBLUP-equivalent) achieved $R^2 = 0.646$ and Pearson $r = 0.824$, validating genomic prediction as a viable approach for DTF in this panel and setting a quantitative benchmark for future model improvements.
- SHAP analysis identified chromosome 5 SNPs as the most predictively important features in the non-linear model, revealing biological signal not captured by the linear GWAS - demonstrating the complementary value of ML explainability methods.
- A genomic inflation factor of $\lambda = 1.427$ signals the need for mixed-model GWAS in the full analysis. This is a direct, actionable outcome of the prototype that prevents a methodological error in the main PhD analyses.
- Three dual-method candidate loci (top-50 overlap between GWAS and SHAP) were identified as highest-confidence targets for experimental follow-up and transcriptomic integration.

The immediate next steps are: (1) re-run GWAS with the full 539,643 SNP set; (2) implement mixed-model GWAS with kinship matrix; (3) integrate RNA-seq transcriptomics to enable eQTL analysis at the significant loci; (4) extend to single-cell resolution to identify cell-type-specific regulatory mechanisms; and (5) incorporate structural variants from long-read sequencing following the pangenome approach of Golicz et al. (2025). Each of these extensions builds directly on validated components of the prototype pipeline, ensuring a rigorous and reproducible path from this starter analysis to publication-ready science.

References

- Bornhofen E. et al. (2024). Genotypic and phenotypic data for the ProFaba diverse panel of faba bean accessions. Zenodo. <https://doi.org/10.5281/zenodo.13908861>
- Jayakodi M., Golicz A.A. et al. (2023). The giant diploid faba genome unlocks variation in a global protein crop. *Nature* 615, 652–659. <https://doi.org/10.1038/s41586-023-05791-5>
- Golicz A.A. et al. (2025). Structural variants in *Brassica napus* pangenomes affect gene expression and eQTL analysis. *Theoretical and Applied Genetics*.
- Lundberg S.M. & Lee S.I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems* 30.
- Lippolis R. et al. (2025). GWAS for agronomic and seed quality traits in the faba bean CGN diversity panel. *BMC Plant Biology*.